

Software Requirements Specification (SRS)

Project Title

StudyMate – Student Learning and Academic Management System

1. Introduction

1.1 Purpose

The purpose of StudyMate is to provide students, teachers, and administrators with one platform for managing academic activities. The system will make it easier to access courses, notes, assignments, quizzes, attendance, timetable, marks, and other academic information.

This SRS document describes the main requirements and features of the StudyMate system.

1.2 Scope

StudyMate is an academic management system designed for educational institutions. It will provide different features according to the user's role.

The system will mainly support three types of users:

- **Student**
- **Teacher**
- **Admin**

Students will be able to manage their academic activities, view learning materials, submit assignments, attempt quizzes, check attendance, and view marks.

Teachers will be able to manage courses, upload materials, create assignments and quizzes, record attendance, and manage student marks.

Administrators will manage users, departments, courses, and other system information.

1.3 Definitions and Abbreviations

Term	Meaning
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SRS	Software Requirements Specification
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Admin	System Administrator
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Student	User who uses the system for academic activities
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Teacher	User who manages courses and student activities
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GPA	Grade Point Average
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UI	User Interface
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1.4 References

The SRS is prepared according to the general structure and guidelines of the **IEEE Software Requirements Specification format**.

1.5 Overview

This document describes the overall system, users, functional requirements, non-functional requirements, and external interfaces of StudyMate.

2. Overall Description

2.1 Product Perspective

StudyMate will work as a centralized academic management system. Instead of using different platforms for notes, assignments, attendance, marks, and timetables, users will be able to access these academic activities from one system.

2.2 Product Functions

The main functions of StudyMate are:

- User registration and login
- Role-based access
- Course management
- Notes and study material

- Assignment management
- Quiz management
- Attendance management
- Timetable management
- Marks and GPA information
- Notifications
- Student progress tracking
- User and department management

2.3 User Classes and Characteristics

Student

Students will use the system to:

- View courses
- Access notes
- Submit assignments
- Attempt quizzes
- Check attendance
- View timetable
- Check marks and progress

Teacher

Teachers will use the system to:

- Manage courses
- Upload notes
- Create assignments
- Create quizzes
- Record attendance

- Enter marks
- View student progress

Admin

Administrators will use the system to:

- Manage users
- Manage departments
- Manage courses
- Manage system information

2.4 Operating Environment

The system may be developed as a web or mobile-based application. It should work on commonly used devices such as computers, laptops, tablets, and smartphones.

2.5 Design and Implementation Constraints

- The system should have a simple and user-friendly interface.
- Users should only access features allowed for their role.
- The system should require an internet connection for online features.
- The system should use a suitable database for storing academic information.

2.6 Assumptions and Dependencies

- Users will have valid login information.
- Teachers and administrators will provide correct academic information.
- The system will depend on a database for storing and retrieving information.
- Users will have access to a suitable device and internet connection.

3. Specific Requirements

3.1 Functional Requirements

We will use **25 functional requirements**.

FR-01 – User Registration

The system shall allow new users to create an account.

FR-02 – User Login

The system shall allow users to log in using their account credentials.

FR-03 – Role-Based Access

The system shall provide different features according to the user's role.

FR-04 – Profile Management

The system shall allow users to view and update their basic profile information.

FR-05 – Course Viewing

The system shall allow students to view their enrolled courses.

FR-06 – Course Management

The system shall allow teachers and admins to add, update, and manage courses.

FR-07 – Study Materials

The system shall allow teachers to upload notes and other study materials.

FR-08 – Material Access

The system shall allow students to view and access available study materials.

FR-09 – Assignment Creation

The system shall allow teachers to create and publish assignments.

FR-10 – Assignment Submission

The system shall allow students to submit their assignments.

FR-11 – Assignment Status

The system shall allow students to check the status of their submitted assignments.

FR-12 – Quiz Creation

The system shall allow teachers to create and manage quizzes.

FR-13 – Quiz Attempt

The system shall allow students to attempt available quizzes.

FR-14 – Quiz Result

The system shall display quiz results to students.

FR-15 – Attendance Recording

The system shall allow teachers to record student attendance.

FR-16 – Attendance Viewing

The system shall allow students to view their attendance record.

FR-17 – Timetable

The system shall allow students and teachers to view their timetable.

FR-18 – Marks Management

The system shall allow teachers to enter and update student marks.

FR-19 – Marks Viewing

The system shall allow students to view their marks.

FR-20 – GPA and Progress

The system shall allow students to view their GPA and academic progress.

FR-21 – Notifications

The system shall send notifications about important academic activities.

FR-22 – Student Progress

The system shall allow teachers to view the academic progress of students.

FR-23 – User Management

The system shall allow administrators to add, update, and manage users.

FR-24 – Department Management

The system shall allow administrators to manage departments and related information.

FR-25 – Search

The system shall allow users to search for available academic information.

3.2 Non-Functional Requirements

NFR-01 – Usability

The system should have a simple and easy-to-use interface.

NFR-02 – Performance

The system should respond to normal user requests within a reasonable time.

NFR-03 – Security

The system should protect user accounts and academic information from unauthorized access.

NFR-04 – Reliability

The system should provide accurate and consistent information during normal use.

NFR-05 – Maintainability

The system should be designed so that new features and changes can be added easily.

4. External Interface Requirements

4.1 User Interface

The system should provide a simple and clear interface. Users should be able to easily navigate between courses, assignments, quizzes, attendance, timetable, marks, and other features.

4.2 Hardware Interface

The system should work on commonly used computers, laptops, tablets, and smartphones.

4.3 Software Interface

The system will require a suitable database and software environment to store and manage user and academic information.

4.4 Communication Interface

The system will require an internet connection for communication between the application and the server/database.

5. Other Requirements

The system should be designed in a way that allows future features to be added. For example, StudyMate may later support online classes, fee information, parent access, or additional academic reports.

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